

Exercise: Suppose that random variable X has pmf

$$f_X(k) = \frac{1}{2k^2 z(2)}, \quad k = \pm 1, \pm 2, \dots$$

What is $E(X)$ in this case?

Technical assumption: In order for $E(X)$ to exist,

$$E(X) = \underbrace{\sum_{\substack{k \geq 0: \\ f_X(k) > 0}} k f_X(k)}_{(1)} + \underbrace{\sum_{\substack{k < 0: \\ f_X(k) > 0}} k f_X(k)}_{(2)}$$

need either (1) or (2) to be finite.

Cannot end up with $\infty - \infty$.

In this example (1) and (2) are infinite, so $E(X)$ does not exist.

e.g. $E(X+Y) = E(X) + E(Y)$ is often presented as always being true, but what if $Y = -X$, $E(X) = \infty$

$$1 + 2 + 3 + \dots = -\frac{1}{12}$$

Rule for the Lazy Statistician

It is possible to prove that

$$E(h(X)) = \sum_{k: f_X(k) > 0} h(k) f_X(k)$$

This result, sometimes called **the Rule for the Lazy Statistician**, is a **theorem**, and not a **definition**.

Exercise: What would the **definition** of $E(h(X))$ be?

Let $Y = h(X)$, ie. $Y(\omega) = h(X(\omega))$

Derive the pmf for Y , f_Y

$$E(Y) = \sum_{k: f_Y(k) > 0} k f_Y(k)$$

The Rule allows us to avoid deriving f_Y

An important case is the **variance** of a random variable, defined to be

$$V(X) = E\left((X - E(X))^2\right) = E(X^2) - E(X)^2.$$

The **standard deviation** is the square root of the variance.

“Named” Discrete Distributions

Some distributions are so widely utilized that they have “names.” Keep in mind, however, that these are simply special cases of the general class of discrete distributions.

The distributions we will consider here are as follows:

1. The Bernoulli(p) Distribution
2. The Binomial(n, p) Distribution
3. The Geometric(p) Distribution
4. The Poisson(λ) Distribution

In each case, there are **parameters** that specify the exact form of the distribution. So, in fact, there is not “a Binomial Distribution,” but instead a **“family of Binomial Distributions.”**

1 The Bernoulli(p) Distribution

2 A **Bernoulli Trial** is an experiment that has two possible outcomes (typi-
3 cally called "success" and "failure.")

4 X is said to have the **Bernoulli(p) distribution** if $X = 1$ when the trial is a
5 "success" and $X = 0$ when it is a "failure," and $P(X = 1) = p$.

6 **Exercise:** What is the pmf of X in this case? What are $E(X)$ and $V(X)$?

$$f_X(k) = \begin{cases} p, & k=1 \\ 1-p, & k=0 \end{cases}$$

$$E(X) = p$$

$$\begin{aligned} V(X) &= E(X^2) - E(X)^2 \\ &= p(1-p) \end{aligned}$$

1 The Binomial(n, p) Distribution

2 Consider a sequence of n independent Bernoulli trials, and for each trial
3 $P(\text{success}) = p$. If X is the number of “successes,” in those n trials, then X
4 has the **binomial(n, p) distribution**.

5 **Example:** If I roll a die ten times, and X equals the number of “1”'s, then
6 we write $X \sim \text{binomial}(n = 10, p = 1/6)$. In this case, “success” is defined
7 as “rolling a ‘1’.”

8 If $X \sim \text{binomial}(n, p)$, then

$$9 \quad f_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n$$

10 and $E(X) = np$ and $V(X) = np(1-p)$.

Exercise: A life insurance company is interested in the number of claims that it will receive on a particular block of life insurance policies in the coming year. All the policies in the block are written on lives of people who are currently age 65. There are presently 100 active policies in the block. (None of the policies in the block are written on more than one life, and no insured person is covered by more than one policy.). Based on the United States Life Tables the company estimates that the probability of any given 65-year-old dying in the next year is 0.0145 (In actuarial notations, $q_{65} = 0.0145$). Assume that the future lifetime of the insured persons are independent.

Determine the probability that the company receives more than two claims on this block of 100 policies in the coming year.

$X = \# \text{ of claims}$

$X \sim \text{binomial}(n=100, p=0.0145)$

← $P(\text{success on one trial})$

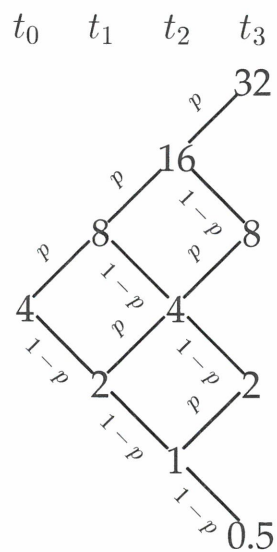
$$P(X > 2) = 1 - F_X(2)$$

In Python: `import scipy.stats as sc`

`1 - sc.binom.cdf(2, 100, 0.0145)`

$$= 0.178$$

Exercise: The **binomial model** for stock prices describes the evolution of prices in discrete time. Starting at a value S_0 the value in the next period could be uS with probability p or dS with probability $1 - p$. Assume that $u > d$. It is often assumed that $u = 1/d$, in which case a simple tree can be used to illustrate the possible paths. For example, if $S_0 = 4$, $u = 2$, and $d = 1/2$, then the model has the following structure:



Let S_t denote the value at time t . Describe the distribution of S_t .

Assume start at time $t=0$, future times are $t=1, 2, \dots$

Then, S_t has $t+1$ possible values
So, S_t is a discrete RV

The possible values can be written as

$$S_0 U^k D^{t-k}, \quad k=0, 1, \dots, t$$

for S_t . Here, $k = \#$ of up steps.

$$P(S_t = S_0 U^k D^{t-k}) = P(\text{exactly } k \text{ up steps})$$

$$= \binom{t}{k} p^k (1-p)^{t-k}, \quad k=0, 1, \dots, t$$

since $\#$ of up steps has binomial(t, p) dist.
Note we are assuming steps are independent.

S_t does not have a binomial dist.

Comment: $S_t = S_0 X_1 X_2 \dots X_t$ where $X_1, \dots, X_t$

are independent RVs, each is such that

$$P(X_i = U) = p \quad P(X_i = D) = 1-p$$

$$S_0, \log S_t = \log S_0 + \underbrace{\sum_{i=1}^t \log X_i}_{\text{sum of independent, identically dist RVs}}$$

As $t \rightarrow \infty$, $\log S_t$ will converge to a normal dist. by CLT

From Stochastic Calculus for Finance I, Shreve

2.2 Random Variables, Distributions, and Expectations

A random experiment generally generates numerical data. This gives rise to the concept of a random variable.

Definition 2.2.1. Let $(\Omega, \mathbb{P})$ be a finite probability space. A random variable is a real-valued function defined on Ω . (We sometimes also permit a random variable to take the values $+\infty$ and $-\infty$.)

Example 2.2.2 (Stock prices). Recall the space Ω of three independent coin-tosses (2.1.1). As in Figure 1.2.2 of Chapter 1, let us define stock prices by the formulas

$$S_0(\omega_1\omega_2\omega_3) = 4 \text{ for all } \omega_1\omega_2\omega_3 \in \Omega_3,$$

$$S_1(\omega_1\omega_2\omega_3) = \begin{cases} 8 & \text{if } \omega_1 = H, \\ 2 & \text{if } \omega_1 = T, \end{cases}$$

$$S_2(\omega_1\omega_2\omega_3) = \begin{cases} 16 & \text{if } \omega_1 = \omega_2 = H, \\ 4 & \text{if } \omega_1 \neq \omega_2, \\ 1 & \text{if } \omega_1 = \omega_2 = T, \end{cases}$$

$$S_3(\omega_1\omega_2\omega_3) = \begin{cases} 32 & \text{if } \omega_1 = \omega_2 = \omega_3 = H, \\ 8 & \text{if there are two heads and one tail,} \\ 2 & \text{if there is one head and two tails,} \\ .50 & \text{if } \omega_1 = \omega_2 = \omega_3 = T. \end{cases}$$

Here we have written the arguments of S_0 , S_1 , S_2 , and S_3 as $\omega_1\omega_2\omega_3$, even though some of these random variables do not depend on all the coin tosses. In particular, S_0 is actually not random because it takes the value 4, regardless of how the coin tosses turn out; such a random variable is sometimes called a *degenerate random variable*. $\square$

It is customary to write the argument of random variables as ω , even when ω is a sequence such as $\omega = \omega_1\omega_2\omega_3$. We shall use these two notations interchangeably. It is even more common to write random variables without any arguments; we shall switch to that practice presently, writing S_3 , for example, rather than $S_3(\omega_1\omega_2\omega_3)$ or $S_3(\omega)$.

According to Definition 2.2.1, a random variable is a function that maps a sample space Ω to the real numbers. The *distribution* of a random variable is a specification of the probabilities that the random variable takes various values. A random variable is not a distribution, and a distribution is not a random variable. This is an important point when we later switch between the actual probability measure, which one would estimate from historical data, and the risk-neutral probability measure. The change of measure will change

distributions of random variables but not the random variables themselves. We make this distinction clear with the following example.

Example 2.2.3. Toss a coin three times, so the set of possible outcomes is

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

Define the random variables

$$X = \text{Total number of heads}, \quad Y = \text{Total number of tails}.$$

In symbols,

$$\begin{aligned} X(HHH) &= 3, \\ X(HHT) &= X(HTH) = X(THH) = 2, \\ X(HTT) &= X(THT) = X(TTH) = 1, \\ X(TTT) &= 0, \\ Y(TTT) &= 3, \\ Y(TTH) &= Y(THT) = Y(HTT) = 2, \\ Y(THH) &= Y(HTH) = Y(HHT) = 1, \\ Y(HHH) &= 0. \end{aligned}$$

We do not need to know probabilities of various outcomes in order to specify these random variables. However, once we specify a probability measure on Ω , we can determine the distributions of X and Y . For example, if we specify the probability measure $\tilde{\mathbb{P}}$ under which the probability of head on each toss is $\frac{1}{2}$ and the probability of each element in Ω is $\frac{1}{8}$, then

$$\begin{aligned} \tilde{\mathbb{P}}\{\omega \in \Omega; X(\omega) = 0\} &= \tilde{\mathbb{P}}\{TTT\} = \frac{1}{8}, \\ \tilde{\mathbb{P}}\{\omega \in \Omega; X(\omega) = 1\} &= \tilde{\mathbb{P}}\{HTT, THT, TTH\} = \frac{3}{8}, \\ \tilde{\mathbb{P}}\{\omega \in \Omega; X(\omega) = 2\} &= \tilde{\mathbb{P}}\{HHT, HTH, THH\} = \frac{3}{8}, \\ \tilde{\mathbb{P}}\{\omega \in \Omega; X(\omega) = 3\} &= \tilde{\mathbb{P}}\{HHH\} = \frac{1}{8}. \end{aligned}$$

We shorten the cumbersome notation $\tilde{\mathbb{P}}\{\omega \in \Omega; X(\omega) = j\}$ to simply $\tilde{\mathbb{P}}\{X = j\}$. It is helpful to remember, however, that the notation $\tilde{\mathbb{P}}\{X = j\}$ refers to the probability of a subset of Ω , the set of elements ω for which $X(\omega) = j$. Under $\tilde{\mathbb{P}}$, the probability that X takes the four values 0, 1, 2, and 3 are

$$\begin{aligned} \tilde{\mathbb{P}}\{X = 0\} &= \frac{1}{8}, \quad \tilde{\mathbb{P}}\{X = 1\} = \frac{3}{8}, \\ \tilde{\mathbb{P}}\{X = 2\} &= \frac{3}{8}, \quad \tilde{\mathbb{P}}\{X = 3\} = \frac{1}{8}. \end{aligned}$$

This table of probabilities where X takes its various values records the *distribution* of X under $\tilde{\mathbb{P}}$.

The random variable Y is different from X because it counts tails rather than heads. However, under $\tilde{\mathbb{P}}$, the distribution of Y is the same as the distribution of X :

$$\begin{aligned}\tilde{\mathbb{P}}\{Y = 0\} &= \frac{1}{8}, \quad \tilde{\mathbb{P}}\{Y = 1\} = \frac{3}{8}, \\ \tilde{\mathbb{P}}\{Y = 2\} &= \frac{3}{8}, \quad \tilde{\mathbb{P}}\{Y = 3\} = \frac{1}{8}.\end{aligned}$$

The point here is that the random variable is a function defined on Ω , whereas its distribution is a tabulation of probabilities that the random variable takes various values. A random variable is not a distribution.

Suppose, moreover, that we choose a probability measure $\mathbb{P}$ for Ω that corresponds to a $\frac{2}{3}$ probability of head on each toss and a $\frac{1}{3}$ probability of tail. Then

$$\begin{aligned}\mathbb{P}\{X = 0\} &= \frac{1}{27}, \quad \mathbb{P}\{X = 1\} = \frac{6}{27}, \\ \mathbb{P}\{X = 2\} &= \frac{12}{27}, \quad \mathbb{P}\{X = 3\} = \frac{8}{27}.\end{aligned}$$

The random variable X has a different distribution under $\mathbb{P}$ than under $\tilde{\mathbb{P}}$. It is the same random variable, counting the total number of heads, regardless of the probability measure used to determine its distribution. This is the situation we encounter later when we consider an asset price under both the actual and the risk-neutral probability measures.

Incidentally, although they have the same distribution under $\tilde{\mathbb{P}}$, the random variables X and Y have different distributions under $\mathbb{P}$. Indeed,

$$\begin{aligned}\mathbb{P}\{Y = 0\} &= \frac{8}{27}, \quad \mathbb{P}\{Y = 1\} = \frac{12}{27}, \\ \mathbb{P}\{Y = 2\} &= \frac{6}{27}, \quad \mathbb{P}\{Y = 3\} = \frac{1}{27}.\end{aligned} \quad \square$$

Definition 2.2.4. Let X be a random variable defined on a finite probability space $(\Omega, \mathbb{P})$. The expectation (or expected value) of X is defined to be

$$\mathbb{E}X = \sum_{\omega \in \Omega} X(\omega) \mathbb{P}(\omega).$$

When we compute the expectation using the risk-neutral probability measure $\tilde{\mathbb{P}}$, we use the notation

$$\tilde{\mathbb{E}}X = \sum_{\omega \in \Omega} X(\omega) \tilde{\mathbb{P}}(\omega).$$

The variance of X is

$$\text{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}X)^2\right].$$

Geometric(p) Distribution

Again, consider a sequence of independent Bernoulli trials, each with probability of “success” equal to p . Let X denote the number of trials **prior to** the first “success.” Then $X \sim \text{geometric}(p)$.

In other words, X is the number of “failures” prior to the first success.¹

The pmf in this case is

$$f_X(k) = (1 - p)^k p, \quad k = 0, 1, 2, \dots$$

Also, $E(X) = (1 - p)/p$ and $V(X) = (1 - p)/p^2$. (Assumes $p > 0$.)

Note: Remember that when $|r| < 1$,

$$\sum_{k=a}^{\infty} r^k = \frac{r^a}{1 - r}.$$

¹Note that in some texts this distribution is defined as the number of trials **up to and including** the first success.

- 1 **Exercise (From HOTS):** You have three children, but only one apple. You
2 want to toss a fair coin to determine which child gets the apple. You want
3 each child to be equally likely to get the apple. What is your strategy?
4 What is the expected number of tosses needed to complete this strategy?

5 Flip twice

6 HH $\Rightarrow$ child 1

7 HT $\Rightarrow$ " 2

8 TH $\Rightarrow$ " 3

9 TT $\Rightarrow$ redo

10 A "trial" consists of a pair of flips.

11 A "success" is when a child wins the apple.

12 $X = \#$ of failed trials $= \#$ of "redo"

13 $X \sim \text{Geometric}(p = 3/4)$

14 Number of flips $= 2X + 2$

15
$$E(\# \text{ of flips}) = 2E(X) + 2$$

16
$$= 2\left(\frac{1-p}{p}\right) + 2 = \frac{8}{3}$$

Exercise: Three people (call them A, B, and C) roll a die in that order. They repeat this order until the first "6," at which time that person wins the game. What is the probability that A wins? What is the probability that B wins?

$X = \#$ of tosses before first "6"
 $X \sim \text{Geometric}(p = 1/6)$

$$P(\text{A wins}) = P(X=0 \text{ or } X=3 \text{ or } X=6 \dots)$$

$$= \sum_{k=0}^{\infty} P(X=3k)$$

$$= \sum_{k=0}^{\infty} \left(1 - \frac{1}{6}\right)^{3k} \left(\frac{1}{6}\right) \quad \text{geometric pmf}$$

$$= \frac{1}{6} \sum_{k=0}^{\infty} \left(\left(\frac{5}{6}\right)^3\right)^k$$

$$= \frac{1}{6} \left(\frac{1}{1 - \left(\frac{5}{6}\right)^3} \right) \quad \leftarrow \text{using geometric series result}$$

$$= \frac{36}{91}$$

$$P(\text{B wins}) = P(X=1 \text{ or } X=4 \text{ or } \dots)$$

$$\text{Or: } P(B \text{ wins}) = P(B \text{ wins} \mid \text{first roll is } 6) P(\text{first is } 6) \\ + P(B \text{ wins} \mid \text{first roll not } 6) P(\text{first not } 6)$$

LFTP

$$= (0) \left(\frac{1}{6}\right) + \left(\frac{36}{91}\right) \left(\frac{5}{6}\right) = \frac{30}{91}$$

$$P(C \text{ wins}) = \left(\frac{5}{6}\right) \left(\frac{30}{91}\right) =$$

$$\text{Or: } P(B \text{ wins}) = \left(\frac{5}{6}\right) P(A \text{ wins}) \\ P(C \text{ wins}) = \left(\frac{5}{6}\right) P(B \text{ wins}) = \left(\frac{25}{36}\right) P(A \text{ wins})$$

$$\text{and } P(A \text{ wins}) + P(B \text{ wins}) + P(C \text{ wins}) = 1$$

Poisson(λ) Distribution

A random variable X has the **Poisson(λ) Distribution**, for $\lambda > 0$, if it has pmf

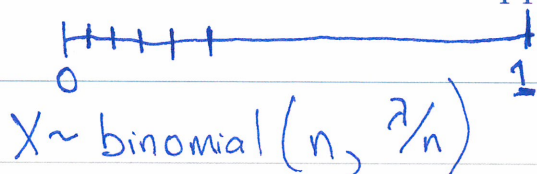
$$f_X(k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

This is useful for modelling counts, especially the number of times something happens in a fixed period of time. It holds that $E(X) = \lambda$ and $V(X) = \lambda$.

The Poisson pmf can be derived via the following considerations: Fix some $\lambda > 0$, and let $X \sim \text{binomial}(n, p = \lambda/n)$ and let $Y \sim \text{Poisson}(\lambda)$. Then for all $k = 0, 1, \dots$, $n > \lambda$

$$\lim_{n \rightarrow \infty} P(X = k) = P(Y = k).$$

Exercise: Divide a fixed time interval into n equally-sized segments. For each of these subintervals, place a “mark” in it with probability λ/n . (Fix $\lambda > 0$, and choose $n > \lambda$.) Let X equal the total number of “marks.” What is the distribution of X ? What happens as n grows?



As $n \rightarrow \infty$, dist. of X converges to $\text{Poisson}(\lambda)$,
from result above

$$P(X=k) = \binom{\frac{n!}{k!(n-k)!}}{\left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}}$$

$$= \left(\frac{\lambda^k}{k!}\right) \left(\frac{n!}{(n-k)! n^k}\right) \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-k}$$

as $n \rightarrow \infty$

$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
$\left(\frac{\lambda^k}{k!}\right)$	$\underline{1}$	$e^{-\lambda}$	$\underline{1}$

$$= P(Y=k)$$

Exercise: An auto insurance company has determined that the average number of claims against the comprehensive coverage of a policy is 0.6 per year.

1. What is the probability that a policy holder will file one claim in a year?

2. What is the probability that a policy holder will file more than one claim in a year?

If I'm willing to assume that claims are made independently, then
 $X \sim \text{Poisson}(\lambda = 0.6)$

$$\textcircled{1} \quad P(X=1) = \frac{e^{-\lambda} \lambda^1}{1!} = 0.33$$

$$\textcircled{2} \quad P(X > 1) = 1 - F_X(1)$$

Independent Discrete Random Variables

The notion of independence extends naturally to discrete random variables. In particular, for a pair of discrete random variables X and Y , they are said to be **independent** if for all a and b ,

$$P(X = a \text{ and } Y = b) = P(X = a)P(Y = b).$$

Exercise: Suppose that X has the $\text{Poisson}(\lambda_1)$ distribution, and that Y has the $\text{Poisson}(\lambda_2)$ distribution. Does $X + Y$ have the Poisson distribution?

For any $k=0,1,2,\dots$

$$P(X+Y=k) = \sum_{i=0}^k P(X=i \text{ and } Y=k-i)$$

$$= \sum_{i=0}^k P(X=i) P(Y=k-i) \quad \text{Assuming } X, Y \text{ ind.}$$

$$= \sum_{i=0}^k \left(\frac{e^{-\lambda_1} \lambda_1^i}{i!} \right) \left(\frac{e^{-\lambda_2} \lambda_2^{k-i}}{(k-i)!} \right)$$

$$= \frac{e^{-(\lambda_1+\lambda_2)}}{k!} \sum_{i=0}^k \left(\frac{k!}{i!(k-i)!} \right) \lambda_1^i \lambda_2^{k-i}$$

$$= \frac{e^{-(\lambda_1+\lambda_2)}}{k!} (\lambda_1 + \lambda_2)^k \quad \text{by binomial theorem}$$

So, $X+Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$ assuming X, Y ind.

Continuous Distributions

Unlike in the discrete case, **continuous distribution** can assume any value in an interval on the real line (bounded or unbounded).

To describe a continuous distribution of random variable X , we often use the **probability density function (pdf)**, or just “density,” denoted f_X .

The density f_X has the following properties:

1. $f_X(x) \geq 0$ for all x

2.

$$\int_{-\infty}^{\infty} f_X(u) du = 1$$

3. For any $a \leq b$,

$$P(a \leq X \leq b) = \int_a^b f_X(u) du$$

4. The expected value of X is calculated as

$$E(X) = \int_{-\infty}^{\infty} u f_X(u) du$$

Comments

1. The set of values x for which $f_X(x) > 0$ is called the **support** for the distribution of the random variable X .

2. Note that Condition 3 above implies that $P(X = a) = 0$ for all $a \in \mathbb{R}$.

Hence,

$$P(a < X \leq b) = P(a \leq X < b) = P(a < X < b) = P(a \leq X \leq b).$$

3. The CDF is defined as above, and all of the properties given above hold. Note, however, that since $P(X = a) = 0$ for all a , then $F_X(x)$ must be continuous. Also, from the Fundamental Theorem of Calculus,

$$f_X(x) = \frac{\partial}{\partial x} F_X(x).$$

4. In order for $E(X)$ to exist², need either

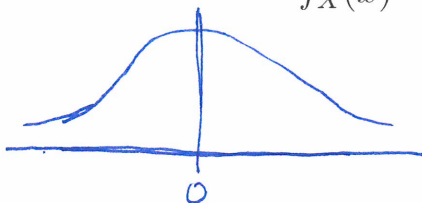
$$\int_0^\infty u f_X(u) du < \infty$$

or

$$\int_{-\infty}^0 u f_X(u) du > -\infty.$$

Otherwise, one is faced with defining $\infty - \infty$. The Cauchy Distribution is a classic example for which $E(X)$ does not exist. In that case,

$$f_X(x) = \frac{1}{\pi(1+x^2)}, \quad -\infty < x < \infty.$$



²We stress that " $E(X)$ exists" is not the same as " $E(X)$ is finite." If $E(X) = \infty$, then $E(X)$ does exist.

1 Rule for the Lazy Statistician

2 The Rule for the Lazy Statistician in the continuous case is

$$3 \quad E(g(X)) = \int_{-\infty}^{\infty} g(u) f_X(u) du.$$

4 Again, it is worth stressing that this is **not** the definition of $E(g(X))$. It is
5 instead a **theorem** that builds on some sophisticated probability theory.

6 To **define** $E(g(X))$, one could start by letting $Y = g(X)$ and then derive the
7 pdf for Y , call it f_Y , and then

$$8 \quad E(g(X)) = E(Y) = \int_{-\infty}^{\infty} u f_Y(u) du.$$

9 Hence, one is “lazy” in that it is possible to find $E(g(X))$ without deriving
10 f_Y . **But**, in order to calculate probabilities of the form

$$11 \quad P(a \leq g(X) \leq b) = P(a \leq Y \leq b)$$

12 one would need to derive f_Y .

Exercise: The distribution function (CDF) for a continuous random variable X is given by

$$F_X(x) = \begin{cases} 1 - e^{-kx^2} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

~~$$F_X(x) = 1 - e^{-kx^2}$$~~

where k is a positive constant.

1. Determine the density function $f_X(x)$.

2. Show that regardless of the value of $k > 0$ this is a legitimate density function.

3. For $k = 1$, find $P(2 < X < 4)$ and $P(X > 2)$.

4. Find $E(X)$ for any value of k .

5. Find $E(1/X)$ for any value of k .

$$\textcircled{1} f_X(x) = \frac{d}{dx} F_X(x) = \begin{cases} 2xk e^{-kx^2}, & x \geq 0 \\ 0 & , x < 0 \end{cases}$$

$$\textcircled{2} \text{ Show that } f_X(x) \geq 0 \text{ for all } x \\ \text{and that } \int_{-\infty}^{\infty} f_X(u) du = 1$$

$$\textcircled{3} k=1$$

$$P(2 < X < 4) = \int_2^4 f_X(u) du = \int_2^4 2u e^{-u^2} du = 0.018$$

$$\text{Or: } P(2 < X < 4) = F_X(4) - F_X(2)$$

$$\textcircled{4} \quad E(X) = \int_{-\infty}^{\infty} u f_X(u) du$$

$$= \int_0^{\infty} u 2uk e^{-ku^2} du$$

$$= 2k \int_0^{\infty} u^2 e^{-ku^2} du$$

$$= 2k \left(\frac{\Gamma\left(\frac{3}{2}\right)}{2k^{3/2}} \right)$$

Using integral
table, 2nd row
3rd column
 $m=2, a=k$

$$= \frac{\sqrt{\pi}}{2\sqrt{k}}$$

$$\textcircled{5} \quad E\left(\frac{1}{X}\right) \neq \frac{1}{E(X)}$$

$$E\left(\frac{1}{X}\right) = \int_{-\infty}^{\infty} \frac{1}{u} f_X(u) du \quad \text{Rule for Lazy Stat.}$$

$$= \int_0^{\infty} 2k e^{-ku^2} du$$

$$= 2k \int_0^{\infty} e^{-ku^2} du$$

$$= 2k \left(\frac{1}{2} \frac{\sqrt{\pi}}{\sqrt{k}} \right) \quad \text{from integral table}$$

$$= \sqrt{\pi k}$$

Exercise: The density function for a continuous random variable X is given by:

$$f_X(x) = \begin{cases} 4x & 0 \leq x < \frac{1}{2} \\ 4 - 4x & \frac{1}{2} \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$$

1. Determine a formula for the cdf of X , $F_X(x)$.

2. Determine $P(0.25 < X \leq 0.75)$

3. Find $E(X)$.

“Named” Continuous Distributions

There are a wide range of continuous distributions that are important in Statistics. A few important ones will be presented now, but others will appear later, when needed. The ones shown here:

1. The Uniform(a, b) Distribution

2. The Pareto(α, λ) Distribution

3. The Exponential(λ) Distribution

4. The Gamma(α, β) Distribution

5. The Normal(μ, σ^2) Distribution

1 The Uniform(a, b) Distribution

2 If one says that X has the **Uniform(a, b) distribution**, this means that

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}$$

4 Heuristically, any value between a and b is “equally likely.” (Of course,
5 this is not a technically correct way of thinking about it, as all values have
6 probability zero under a continuous distribution.)

7 It is not difficult to show that $E(X) = (a + b)/2$ and $V(X) = (b - a)^2/12$.

8 Further,

$$F_X(x) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a < x < b \\ 1, & x \geq b \end{cases}$$

- 1 **Exercise:** Take a stick of unit length and break it at random. What is the
 2 distribution of the length of the longer piece?



$$U \sim \text{Uniform}(0, 1)$$

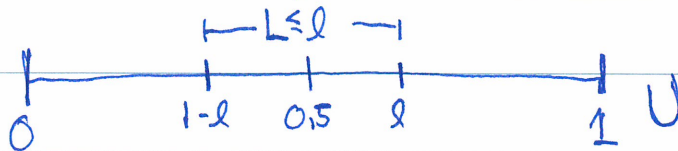
L = Length of longer piece

Derive the CDF of L . know $0.5 \leq L \leq 1$

$$F_L(l) = 0 \quad \text{for } l < 0.5$$

$$F_L(l) = 1 \quad \text{for } l > 1$$

For $0.5 \leq l \leq 1$, when is $L \leq l$? Or, when is $L > l$?



$$P(L \leq l) = P(1-l \leq U \leq l) = 2l - 1$$

(since $U \sim \text{Uniform}(0, 1)$, prob. = length)

$$F_L(l) = \begin{cases} 0, & l < 0.5 \\ 2l - 1, & 0.5 \leq l \leq 1 \\ 1, & l > 1 \end{cases}$$

This matches the CDF of $\text{Uniform}(0.5, 1)$

Hence, $L \sim \text{Uniform}(0.5, 1)$